

eCourts Cause List Scraper

Project Documentation

October 18, 2025

Abstract

This application automates downloading judicial cause lists from the official Indian eCourts portal. A user can specify a state, district, court complex, and date through a simple web interface, and the application will download all available cause list PDFs for that day.

1 How It Works

This project uses **Selenium** to control a live web browser because the eCourts website is **dynamic**. Content on the site is loaded with JavaScript after user interactions (like selecting a state), which a standard library like `requests` cannot handle. Selenium mimics a human user, allowing it to wait for elements to load, select from dropdowns, and type into forms.

Key Automation Techniques Used

- **WebDriverWait:** Makes the script wait patiently for page elements to appear, preventing crashes on slow-loading pages.
- **Dynamic Content Waiting:** A custom lambda function is used to wait until dropdowns are populated with options (e.g., waiting for districts to load after a state is selected). This is a highly reliable method.
- **XPath Searching:** A specific XPath query (`//*[@id='causelist_table']/a[contains('@.pdf')]`) is used to precisely locate and extract all PDF download links from the results table.

2 Quick Start Guide

2.1 Prerequisites

- Python 3.6+
- Google Chrome browser

2.2 Setup & Installation

1. **Organize Files:** Your project folder must be structured as follows:

```
/ecourts_scraper
|-- app.py
|-- scraper.py
|-- README.md
|-- /templates
    |-- index.html
```

2. **Install Libraries:** Open a terminal in your project folder and run:

```
pip install Flask selenium webdriver-manager requests
```

Note: webdriver-manager automatically handles the chromedriver setup.

2.3 Running the Application

1. In your terminal, navigate to the project folder.
2. Run the Flask app:

```
python app.py
```

3. Open your web browser and go to <http://127.0.0.1:5000>.

3 Usage Instructions

1. Fill out the form in your browser.
 - **CRITICAL:** The State, District, and Court Complex names must be an *exact, case-sensitive match* to the names on the eCourts website.
2. Click the "Start Download" button.
3. A confirmation message will appear on the page. The download will run in the background. You can monitor its live progress in your terminal. A new Chrome window will open and close on its own; this is normal.

3.1 Where are the Files Saved?

Downloaded PDFs are saved in a new folder inside your project directory, named with the pattern: Causelists_[District Name]_[Date].

Example: Causelists_New_Delhi_18-10-2025